# KEVIN KWAN

Software Engineer II | Conversational AI & Voice Automation
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## SUMMARY

Software Engineer II specializing in conversational AI and voice automation at GEICO. Build and operate the Sierra AI billing voice agent and Amazon Connect IVR platforms on AWS — 436,112 conversations at ~51% containment (May–Oct 2026), ~30,000 calls per week, and a Proof of Coverage IVR at ~80% containment. Promoted from Software Engineer I in February 2026.

## WORK EXPERIENCE

**GEICO** July 2024 – Present
*Atlanta, GA (Remote)*
**Software Engineer II — Conversational AI & IVR** February 2026 – Present
**Software Engineer I — Technology Development Program** July 2024 – February 2026

- Co-built and operate the Sierra AI billing voice agent (live since May 2026), handling ~30,000 GEICO billing calls per week — payments, rescheduling, AutoPay, billing questions — in English and Spanish; contained ~51% of 436,112 conversations without a live-agent transfer (May–Oct 2026), trending 48% → 52% as weekly volume grew ~10×.
- Delivered end-to-end make-payment for the agent (saved method, pay-now or scheduled, debit authorization, declined-/expired-card recovery, payments on policies pending cancellation): 22,600+ payments at 92–95% authorization success; shipped net-new Pause AutoPay, 82,800 reschedules, and 192,000 info answers at 98% tool success.
- Re-architected the billing agent from regex-/classifier-based routing to fact-based journeys (178 files, ~20,000 net lines removed) and built simulation-based regression quality gates (5-of-5 pass standard) plus automated code-review follow-up, keeping high-velocity releases safe.
- Fixed 86 of 91 production and UT1 billing bugs at the root cause (median 3 days from report to fix); authored 468 PRs — 309 of 559 commits to the billing agent since May, its largest contributor — and 294 production builds across 45 repos; reviewed 653 PRs for 12+ engineers and onboarded two engineers onto the Sierra platform.
- Extensive on-call and production support: release validation, live IVR testing, outage response and rollbacks, post-mortems, and a 41,829-call conversation-experience review that ranked and closed the agent's biggest customer-experience gaps.
- Migrated GEICO's legacy Cisco UCCE and Nuance IVR to Amazon Connect on AWS, engineering Java traffic-redirection logic for gradual live caller cutover alongside legacy endpoint and Amazon Lex pattern migrations.
- Led the Proof of Coverage IVR (~80% containment) and built the AWS self-service payment IVR (~60% containment), owning the Proof of Coverage disaster-recovery rebuild pipeline and delivering a zero-amount-due flow, reworked billing error handling (declined cards and banks retry instead of transferring to an agent), and new card tokenization.
- Designed AWS Lambda functions and Amazon Lex bots (JavaScript code hooks) connecting internal APIs with Amazon Connect contact flows; built asynchronous, high-throughput Lambda orchestration backed by DynamoDB, implemented secure APIs using AES-256 encryption to protect PII and payment data (credit card and bank data), and maintained Terraform pipelines for secure, repeatable deployments.

**Software Engineer Intern — Publicis Sapient (Publicis Groupe)** June 2023 – August 2023
*Atlanta, GA*

- Engineered a WhatsApp AI chatbot for the Travel & Hospitality industry using Google Vertex AI's PaLM LLM, LangChain, MongoDB, and external APIs to deliver live data and recommendations for Atlanta attractions, hotels, and restaurants.
- Architected Python/Flask RESTful APIs bridging MongoDB, the AI chatbot backend, and an interactive map website, automating CI/CD via GitLab and Docker to deploy on Google App Engine and Firebase Hosting.
- Boosted chatbot response accuracy to 91% through rigorous testing and validation, outperforming the Boston team and all other teams in the Atlanta office.

**Software Engineer, Connected & Automated Vehicles — Georgia Tech EcoCAR** January 2022 – May 2023
*EcoCAR (Electric Vehicle) Mobility Challenge | Atlanta, GA*

- Built a camera/sensor state machine on the Vehicle-to-Everything (V2X) subteam enabling the car to make stop/go decisions across traffic-light scenarios, improving fuel efficiency and ride experience.
- Contributed to Georgia Tech's 1st-place finish in the 2022 EcoCAR Mobility Challenge — a $1M collegiate research program sponsored by the U.S. Department of Energy, Argonne National Labs, General Motors, MathWorks, and Intel.

## TECHNICAL SKILLS AND KNOWLEDGE

**Languages:** Java, Python, TypeScript, JavaScript, C#, C++, HTML, CSS, Lua, MATLAB
**Conversational AI & LLM:** Sierra AI, Amazon Connect, Amazon Lex, LangChain, Google Vertex AI (PaLM), OpenAI, Cyara
**AWS & Cloud Infrastructure:** AWS (Lambda, DynamoDB, API Gateway, S3, CloudWatch), Google Cloud Platform (App Engine, Firebase), Microsoft Azure, Cloudflare
**Web, APIs & Data:** REST APIs (AES Encryption), Flask, Node.js, React, Next.js, Tailwind CSS, MySQL, MongoDB, Cloud Firestore
**DevOps & Tooling:** Terraform, Docker, CI/CD, Git, GitHub, GitLab, SonarQube, Cursor, Claude Code

## EDUCATION

**Georgia Institute of Technology — Atlanta, GA** August 2021 – May 2024

- Bachelor of Science in Computer Science, College of Computing; Threads: Intelligence and Media; *summa cum laude* GPA: 3.9
- Zell Miller Scholarship, Graduated with Highest Honors, Dean's List, Faculty Honors